

Development Protocol

Magnetic field and gradient of an axially magnetized cylinder

This document records the engineering protocol followed to develop the two MATLAB programs in this directory:

- **cylinder_magnet.m** — brute-force surface integral via `integral2`
- **cylinder_magnet_analytic.m** — analytical ϕ' -integration + 1-D radial Gauss–Legendre quadrature

1. Problem statement

Compute the magnetic field $\mathbf{B}(x,y,z)$ and its full gradient tensor $\partial B_\alpha / \partial x_\beta$ outside a uniformly axially magnetized cylinder of radius R , length L and magnetization $\mathbf{M} = M_0 \hat{z}$ with $\mu_0 M_0 = 1$ T, axis aligned with the Z -axis.

Required outputs:

- 1-D on-axis profile of B_z for validation;
- 2-D maps of B_x , B_y , B_z and the gradient tensor in an XY plane at a chosen height $z = z_0$ above the magnet;
- visualization of in-plane field direction and gradient direction.

2. Physics formulation

2.1 Scalar magnetic potential

For uniform magnetization the body is replaced by surface magnetic charges with density $\sigma_m = \mathbf{M} \cdot \mathbf{n}$. Only the two flat end faces carry charge for axial magnetization (top: $z' = +L/2$, $\sigma = +M_0$; bottom: $z' = -L/2$, $\sigma = -M_0$).

$$\varphi_m(\mathbf{r}) = \frac{1}{4\pi} \iint_S \frac{\sigma_m(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS'$$

2.2 Field and gradient

Outside the magnet $\mathbf{H} = -\nabla\phi_m$ and $\mathbf{B} = \mu_0 \mathbf{H}$. Both $\mathbf{B}$ and $\partial B_\alpha / \partial x_\beta$ are obtained by differentiating *under the integral sign* analytically:

$$H_\alpha = \frac{\sigma}{4\pi} \iint_S \frac{\alpha - \alpha'}{d^3} dS'$$
$$\frac{\partial H_\alpha}{\partial x_\beta} = \frac{\sigma}{4\pi} \iint_S \left[-\frac{\delta_{\alpha\beta}}{d^3} + \frac{3(\alpha - \alpha')(\beta - \beta')}{d^5} \right] dS'$$

2.3 Validation references

Near / intermediate field: closed form on the cylinder axis ($\rho = 0$):

$$B_z(0, 0, z) = \frac{\mu_0 M_0}{2} \left[\frac{z + L/2}{\sqrt{R^2 + (z + L/2)^2}} - \frac{z - L/2}{\sqrt{R^2 + (z - L/2)^2}} \right]$$

Far field: point-dipole formula with moment $m = M_0 \cdot \pi R^2 L$,

$$\mathbf{B}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{3(\mathbf{m} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}}{r^3}$$

3. Implementation strategy

Two implementations were produced, each a logical step of the same plan.

3.1 Brute-force (cylinder_magnet.m)

- Each of the three field components and each of the five independent gradient-tensor components is computed by `integral2` per observation point on the (x,y) grid.
- Total: $8 \times N^2$ adaptive 2-D integrations.
- Pros: minimal physics derivation, robust.
- Cons: minutes for a 50×50 grid; not useful for design iteration.

3.2 Analytical reduction (cylinder_magnet_analytic.m)

Step 1 — analytical ϕ' integration. The 2π integral over the disc parameter ϕ' is evaluated in closed form; the result is expressed in terms of the complete elliptic integrals $K(m)$ and $E(m)$ with parameter

$$m(r', \rho, \zeta) = \frac{4r'\rho}{(r' + \rho)^2 + \zeta^2}$$

Step 2 — closed-form integrands. The remaining integrand depends only on r' , ρ , ζ :

$$H_z(\rho, \zeta) = \frac{\sigma\zeta}{\pi} \int_0^R \frac{r' E(m)}{[(r' - \rho)^2 + \zeta^2] \sqrt{(r' + \rho)^2 + \zeta^2}} dr'$$

$$H_\rho(\rho, \zeta) = \frac{\sigma}{2\pi\rho} \int_0^R \frac{[(\rho^2 - r'^2 - \zeta^2)E(m) + ((r' - \rho)^2 + \zeta^2)K(m)]}{[(r' - \rho)^2 + \zeta^2] \sqrt{(r' + \rho)^2 + \zeta^2}} dr'$$

Step 3 — vectorized 1-D quadrature. The radial integral on $[0, R]$ is evaluated by a single 96-point Gauss–Legendre rule (Golub–Welsch), broadcast across the entire observation grid as one matrix product.

Step 4 — gradient tensor. Six perturbation evaluations of the analytic field at observation points $x \pm h$, $y \pm h$, $z \pm h$ give the gradient tensor by central differences. The field is quadrature-noiseless, so a step $h = 10^{-6}$ m gives $\sim 10^{-9}$ accuracy.

Speed-up vs brute force: $\approx 10^3$.

4. Numerical-precision protocol

Issue	Mitigation
$K(m \rightarrow 1)$ logarithmic singularity	clamp $m \leq 1 - 10^{-14}$ (rim only; observations stay outside the magnet)
$\rho = 0$ ($1/\rho$ in H_ρ formula)	guard $\max(\rho, 10^{-9} R)$; then force $H_\rho(\rho < 10^{-12} R) = 0$
0/0 in Cartesian projection on axis	mask $n_z = (\rho > \text{eps})$; set $B_\rho = B_\phi = 0$ on axis
Integration singularity at the rim	observation plane lifted by <code>z0add</code> (default 1 mm) above the magnet

5. Validation protocol

Performed each time the analytic core is modified.

1. On-axis comparison at 200 log-spaced points $z \in [b + 10^{-4}, 20 L]$:

- numerical (1-D Gauss–Legendre) vs exact closed form — must agree to relative 10^{-9} over the full range;
- numerical vs dipole far-field — agree only for $z \gg L$; the log-log error plot must asymptote to a $1/r^4$ -like decay of the residual.

2. Divergence check on the 2-D grid: trace of the gradient tensor

$$\nabla \cdot \mathbf{B} = \partial_x B_x + \partial_y B_y + \partial_z B_z$$

plotted alongside the diagonal components. In source-free conditions the resulting map should be numerical noise (■ component magnitudes).

3. Cross-check between programs: at a small set of grid points, the brute-force script and the analytic script must agree to 10^{-8} relative tolerance.

6. Visualization protocol

Figure	Content
1	On-axis B_z and its relative error vs both references (log-log, 2 panels)
2	B_x , B_y as pcolor, B_z as surf, common color limits, axes in cm
3	B_z color map with quiver of normalized in-plane ($B_x/B_z, B_y/B_z$) and magnet outline
4	Five gradient components plus div B (sanity check) on a 2×3 tiled layout
5	Three components of $\nabla B $ obtained from the gradient tensor and B itself

All plots use docked windows, white background, *jet* colormap for the field components, default colormap for the gradients.

7. Project parameters (current run)

R	= 35 mm	cylinder radius
L	= 60 mm	cylinder length
Br	= 1.0 T	remanent flux density
z0add	= 1 mm	distance above the top face for the XY map
N	= 80	XY grid resolution (analytic version)
Nq	= 96	Gauss-Legendre nodes on $[0, R]$
hd	= 1e-6 m	central-difference step for the gradient tensor

8. Reproducibility

Both scripts are self-contained; no toolboxes beyond base MATLAB are required (ellipke ships with the core product). Running each script from a clean session reproduces all figures. The brute-force script is kept in the repository as the slow but transparent reference implementation; the analytic script is the one used for production runs.